



REVIEW ARTICLES

A review of the use of technology in sport coaching: current trends and future directions

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Abstract

Background and purpose

This article aims to provide an overview of the use of technology in sports training, explore current trends, and highlight future directions in this field.

Material and Methods

This article examines technology integration's benefits, challenges, and potential in sports training practice through a comprehensive literature analysis.

Results

This literature review reveals that technology is essential in improving sports training processes and outcomes. Various technological tools and applications have been developed to help coaches analyze athlete performance, monitor physical condition, provide feedback, and make decisions based on objective, real-time data.

Conclusion

The use of technology in sports training offers promising benefits but also presents challenges. To optimize the use of technology in sports training, adequate training, support, and collaboration between coaches, sports scientists, and technology developers are required. Recent trends include the integration of artificial intelligence and machine learning algorithms, utilizing big data analytics, and the development of mobile applications for remote training and athlete monitoring. The future of technology use in sports training involves the continuous development of innovative tools and applications, as well as the exploration of new technologies such as augmented reality and blockchain. Future research needs to focus on evaluating the long-term effectiveness and impact of technology-based training interventions, identifying best practices for implementation, and addressing ethical considerations related to data privacy and athlete welfare.

Key words: technology, sport coaching, review, current trends, future directions



Анотація

Фадлі Іхсан, Ахмад Насрулло, Сігіт Нугрохо, Ріна Юніана. Огляд використання технологій у спортивному тренуванні: поточні тенденції та майбутні напрямки

Обґрунтування і мета

Ця стаття має на меті надати огляд використання технологій у спортивних тренуваннях, вивчити поточні тенденції та висвітлити майбутні напрямки в цій галузі.

Матеріал і методи

Ця стаття розглядає переваги, проблеми та потенціал інтеграції технологій у практиці спортивних тренувань за допомогою комплексного аналізу літератури.

Результати

Цей огляд літератури показує, що технологія має важливе значення для покращення процесів спортивного тренування та результатів. Було розроблено різні технологічні інструменти та додатки, які допомагають тренерам аналізувати продуктивність спортсменів, контролювати фізичний стан, надавати зворотній зв'язок і приймати рішення на основі об'єктивних даних у реальному часі.

Висновки

Використання технологій у спортивному тренуванні пропонує багатообіцяючі переваги, але також створює проблеми. Щоб оптимізувати використання технологій у спортивних тренуваннях, необхідна адекватна підготовка, підтримка та співпраця між тренерами, спортивними науковцями та розробниками технологій. Останні тенденції включають інтеграцію штучного інтелекту та алгоритмів машинного навчання, використання аналітики великих даних, а також розробку мобільних додатків для віддаленого тренування та моніторингу спортсменів. Майбутнє використання технологій у спортивних тренуваннях передбачає постійний розвиток інноваційних інструментів і програм, а також дослідження нових технологій, таких як доповнена реальність і блокчейн. Майбутні дослідження мають бути зосереджені на оцінці довгострокової ефективності та впливу технологічних тренувальних втручань, визначенні найкращих практик для впровадження та розгляді етичних міркувань, пов'язаних із конфіденційністю даних і добробутом спортсменів.

Ключові слова: технологія, спортивний коучинг, огляд, сучасні тенденції, перспективи

Аннотация

Фадли Ихсан, Ахмад Насрулло, Сигит Нугрохо, Рина Юниана. Обзор использования технологий в спортивных тренировках: современные тенденции и будущие направления

Обоснование и цель

Цель этой статьи — дать обзор использования технологий в спортивных тренировках, изучить текущие тенденции и осветить будущие направления в этой области.

Материал и методы

В этой статье рассматриваются преимущества, проблемы и потенциал интеграции технологий в практике спортивных тренировок посредством всестороннего анализа литературы.

Результаты

Этот обзор литературы показывает, что технологии имеют важное значение для улучшения процессов и результатов спортивных тренировок. Были разработаны различные технологические инструменты и приложения, которые помогают тренерам анализировать результаты спортсменов, следить за их физическим состоянием, обеспечивать обратную связь и принимать решения на основе объективных данных в режиме реального времени.

Выводы

Использование технологий в спортивной подготовке дает многообещающие преимущества, но также создает проблемы. Для оптимизации использования технологий в спортивных тренировках необходимы адекватная подготовка, поддержка и сотрудничество между тренерами, спортивными учеными и разработчиками технологий. Последние тенденции включают интеграцию искусственного интеллекта и алгоритмов машинного обучения, использование анализа больших данных, а также разработку мобильных приложений для удаленных тренировок и мониторинга спортсменов. Будущее использования технологий в спортивных тренировках предполагает постоянную разработку инновационных инструментов и приложений, а также исследование новых технологий, таких как дополненная реальность и блокчейн. Будущие исследования должны быть сосредоточены на оценке долгосрочной эффективности и воздействия тренировочных вмешательств на основе технологий, выявлении лучших практик для внедрения и рассмотрении этических соображений, связанных с конфиденциальностью данных и благополучием спортсменов.

Ключевые слова: технологии, спортивный тренинг, обзор, современные тенденции, будущие направления



Introduction

The use of technology in sports has become an increasingly important topic in the modern sports field. In the digital age, sports training increasingly uses technology to improve performance, analyze data, and monitor athlete conditions during training and competition. Therefore, this study aims to review the current trend of technology use in sports training and identify the direction of technology development in sports training.

According to research conducted by [1], In the journal *Frontiers in Psychology*, technology has been used extensively in sports, especially in athlete performance analysis and sports injury management. Technology can assist coaches and medical staff in monitoring athlete performance and prevent injuries. In addition, technology can also help improve the sports fan's experience, such as through live score applications, video streaming, and social media.

However, the use of technology in sports also has some disadvantages. According to research conducted by [2] In the journal *Sustainability*, using technology in sports can increase energy consumption and carbon footprint, especially in sports equipment that requires high electrical power. Therefore, research on developing environmentally friendly technologies in sports is becoming increasingly important.

In the context of technology development in sports training, several studies have shown that the development of artificial intelligence and big data analytics technologies can assist coaches and medical staff analyse athlete performance data and sports injuries more effectively and efficiently. According to research conducted by [3] In the *IEEE Access* journal, big data analytics technology can assist in analyzing athlete performance data more effectively and provide helpful information for coaches and athletes.

Based on the literature review, using technology in sports has advantages and disadvantages and is experiencing rapid development trends. Therefore, this study will discuss the current trend of technology use in sports training and identify the future development direction of technology in sports training.

Material and methods

Research approach and scope

The research approach used in this journal is the literature review method. This method allowed the authors to collect, review, and analyze previous studies on using technology in sports training. This research includes various types of technology used in sports training, such as mobile applications, measurement sensors, data analysis, and wearable devices. The study also involved multiple sports, including team sports, athletics, fitness, and individual sports (Table 1).

Table 1

Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
Sport Type	Irrelevant studies
Coach Population	Invalid research methods
Publication Type	Insufficient data availability
Time Span	Studies with low methodological quality
Language	Studies with unrepresentative samples
Study Type	
Inclusion Criteria	Exclusion Criteria

In the discussion of the table above, the inclusion and exclusion criteria for the literature review on using technology in sports coaching are described. The inclusion criteria included the relevant type of sport, the population of coaches in focus, the type of publication used, the period of the study, the accepted language, and the kind of study taken into account. On the other hand, exclusion criteria included studies that were not relevant to the topic, invalid research methods, insufficient data, low methodological quality, accessibility limitations, and non-representative samples.

This table helps to clarify the criteria used in the selection of relevant studies and also to avoid the inclusion of studies that do not meet the set standards.

Data collection method

Data collection was done through a comprehensive literature search. Searches were



conducted in various databases such as PubMed, Google Scholar, and IEEE Explore using a combination of keywords such as "technology in sport coaching", "sports training technology", and "digital tools for coaching". In addition, a manual review was also conducted on the references contained in the relevant articles that had been found.

Analysis data

Data from the selected literature sources were analyzed systematically. The authors used a thematic approach to research and organized the relevant literature findings. Key findings on the use of technology in sports training were identified and categorized based on related aspects, such as mobile applications for athlete performance monitoring, data analysis for training strategy optimization, and sensors for biomechanical measurements. Then, the data were synthesized and presented as tables, diagrams, or frameworks to clarify the findings and trends.

In this discussion, the researchers explain in detail how the approach, criteria, data collection methods and data analysis were conducted in their literature review on using technology in sports training. This explanation clearly explains how the authors collected and analyzed the relevant literature and the methods used to compile the literature review.

Literature Review

Definition of technology in the context of sports training

Technology in sports training refers to the use of software, hardware, or a combination of both to improve sports performance. [4], The utilization of technology has become an essential aspect of sports coaching to improve athlete performance. Technology in sports coaching refers to using software, hardware or a combination of both to enhance athlete performance. Technology in sports coaching is defined as using various electronic devices, including wearable devices and mobile applications, as collecting and analyze data relating to athletes' physical and physiological performance, as well as their training routines and competitive situations, to improve their performance, [5].

Various types of technology can be used in sports training, such as video analysis, sensor systems,

kinematic measurements, virtual simulations, etc. Technology in sports training includes a wide range of tools and applications, such as video analysis, sensor systems, kinematic measurements, virtual simulations, and so on. These technologies can assist coaches in monitoring and analyzing sports performance and planning more effective and efficient training programs, [6]. [7], The use of technology in sports training can include a wide variety of tools, such as video analysis, sensor systems, kinematic measurements, virtual simulations, and so on.

The use of technology in sports training has evolved along with the development of technology. [8], The use of technology in sports training can include a wide variety of tools, such as video analysis, sensor systems, kinematic measurements, virtual simulations, and so on.

The use of technology in sports training has evolved along with the development of technology. [9], technology in sports training can assist coaches in "making more accurate decisions, improving athletes' skills and physical condition, and preparing athletes for competition more effectively. [10], technology has been used to develop sports training systems that can assist coaches in tracking and analyzing athletes' sports movements using motion sensors and cameras. Such technology can also assist coaches in facilitating communication and providing timely and accurate feedback to their sportspeople.

The use of technology in sports training also has some challenges, such as cost, complexity, and difficulty integrating technology with existing training programs. [11], The main challenges of using technology in sports training are increasing the accessibility of technology, reducing the cost of developing and using technology, and improving technology integration with existing training programs. [12], The main challenges in using technology in sports training are limited human and financial resources, lack of knowledge and understanding of technology among coaches and athletes, and privacy and data security issues related to using such technology.

Classification of technologies used in sports training

Motion analysis technology

Motion analysis technology records, measures and analyses athletes' movements in various sporting situations. This technology can provide



detailed and accurate information about the athlete's movements, which can help coaches to identify and correct movement errors and improve athlete performance. [13], Movement analysis technologies such as motion capture and video analysis can help coaches to improve athletes' movements by providing timely and specific feedback. In addition, these technologies can help coaches objectively measure athlete performance and provide accurate data for evaluation and training planning.

On the other hand, some critics consider that motion analysis technology can be confusing and overwhelm athletes with too much information that is difficult to understand. [14], motion analysis technology should be used wisely, and coaches should consider the athlete's ability and the sport context when using this technology.

Physical measurement technology

Physical measurement technology is used to measure various physical parameters of athletes, such as strength, endurance, speed and flexibility. This technology can help coaches to determine more focused and effective training programs and monitor athletes' progress during training. [15], Physical measurement technology such as strength gauges and VO2max meters can help coaches to identify athletes' physical strengths and weaknesses and develop more specific and compelling training programs.

In addition, this technology can also help coaches to monitor athletes' progress during training and make necessary adjustments to the training program. However, the use of physical measurement technology also has some limitations. [16], Physical measurements only provide a partial picture of an athlete's ability and do not consider the psychological and tactical factors in sporting events. Therefore, physical measurement technology should be combined with performance measurement during the game to get a complete picture of the athlete's ability.

Simulation technology

Simulation technology is used to simulate certain situations and conditions that coaches and athletes can use to prepare for matches. This technology can help athletes to improve their ability to deal with difficult situations and improve their ability to make the right decisions while playing. [17], Simulation technologies such as virtual and augmented reality can provide athletes with a realistic experience in game preparation.

These technologies can also help athletes to develop their ability to make the right decisions in

difficult and complex situations. However, simulation technology also has some limitations. [17], The virtual experience sometimes differs from the experience in a sporting event. [18], Virtual training provides athletes with a simulated environment that can help them improve their technical skills and decision-making abilities. However, it should be noted that virtual experience is not always the same as experience, and the effectiveness of virtual training needs to be verified through honest competition. Therefore, simulation technology should be combined with the expertise of sports competitions to develop athletes' abilities optimally.

Communication and collaboration technology

Communication and collaboration technologies facilitate communication and collaboration between coaches and athletes in game preparation. These technologies can help coaches to provide timely feedback and support the development of more effective training strategies and programs. [19], Communication and collaboration technologies such as video conferencing and web-based applications can facilitate communication between geographically separated coaches and athletes. These technologies can also help coaches to provide timely feedback and support the development of more effective training strategies and programs.

However, communication and collaboration technologies also have some limitations. [20], These technologies only sometimes facilitate social interaction between coaches and athletes. Therefore, communication and collaboration technologies should be combined with in-person social interactions to build strong relationships between coaches and athletes.

Advantages and disadvantages of using technology in sports training

Allows for more accurate and detailed monitoring and analysis of athletes' physical activity and condition.

Technology in sports training allows coaches and athletes to monitor activity and physical condition more accurately and in detail. This can help evaluate athletes' progress and improve the training techniques and strategies applied. Research conducted by [21], Movement monitoring technology can help identify aspects that need improvement in sports techniques, thereby improving athlete performance.

However, the use of monitoring technology



also has some disadvantages. As stated by [22], Excessive use of movement monitoring technology can cause fatigue or injury to athletes due to overdoing the measured movements. Therefore, coaches need to consider the appropriate and wise use of technology.

Can assist in developing more effective and efficient training strategies

The use of technology in sports training can assist coaches in developing more effective and efficient training strategies. This can be achieved by analyzing data and information obtained from monitoring and measurement technology. Research by [23], Data analysis and information from monitoring technology can assist coaches in identifying athletes' strengths and weaknesses and developing appropriate strategies to improve athletes' performance.

However, the use of technology in the development of training strategies also has some disadvantages. As mentioned by [24], The use of technology may lead coaches to overlook other factors that influence athlete performance, such as psychological and social aspects. Therefore, the use of technology should be combined with a holistic approach to sports training.

It enables more effective communication between coaches and, athletes and facilitates the sharing of information and tasks within the team

Technology in sports training can also enable more effective communication between coaches and athletes and facilitate sharing of information and tasks within the team. This can be achieved through apps or digital platforms that allow coaches and athletes to communicate and share information more quickly and efficiently. Research conducted by [25], Digital training apps can improve communication between coaches and athletes and increase athlete engagement within the team.

The use of technology in communication and collaboration also has some drawbacks. As mentioned by [26], The use of technology can affect the social interaction between coaches and athletes, reducing the intimacy in the relationship between the two. Therefore, coaches need to consider the use of technology wisely and keep the social aspects of the coach-athlete relationship in mind.

Can increase athlete motivation and engagement in training, by providing immediate and engaging feedback.

Technology in sports training can also increase athlete motivation and engagement in

activity, by providing immediate and engaging feedback. This can be achieved through monitoring and measurement technologies that provide real-time feedback on athlete performance. Research by [27], Real-time feedback can increase athlete motivation and improve performance in training.

The use of technology in feedback also has some disadvantages. As stated by [28], too much feedback from technology can make athletes overly dependent on technology and neglect the development of core skills in sports. Therefore, coaches need to consider technology in feedback wisely and pay attention to developing core skills required in sports.

Disadvantages of using technology in sports training

It requires high costs to acquire advanced technology and specialized training and more significant time and effort to learn and master the technology.

The use of technology in sports training requires high costs, especially in acquiring the most up-to-date technology. Along with the development of technology, the prices that must be incurred are getting bigger. According to research conducted by [29], In the Sports Medicine journal, the costs required to acquire technology such as motion analysis systems can cost millions of dollars. The cost of specialized training to use this technology must also be considered. This can be an obstacle for teams or athletes who have limited budgets.

In addition to the high cost, using technology also requires more significant time and effort in learning and mastering the technology. Coaches and athletes must first understand how technology can be used in sports training and how the data generated can be used to improve athlete performance. Research conducted by [30], In the International Journal of Performance Analysis in Sport, the use of technology in sports training can take longer to process data and apply the results of its analysis to training. This can hinder training efficiency and take up time that could otherwise be spent on other aspects of training.

It can generate redundant and irrelevant data, thus confusing coaches and athletes.

Technology in sports training can generate redundant and irrelevant data, confusing coaches and athletes. Research conducted by [31], In the journal Sports Biomechanics, some technologies can produce highly detailed and complex data that is difficult for coaches and athletes to understand. This can hinder training effectiveness and worsen athlete



performance.

It can distract from other important aspects of sports training, such as psychological and social aspects.

The use of technology in sports training can distract from other important aspects of sports training, such as psychological and social aspects. Research conducted by [32], In the journal *Frontiers in Sports and Active Living*, technology can provide highly technical information about athlete performance, but it cannot provide information about psychological and social factors that can affect athlete performance. Therefore, technology should be tailored to training needs and pay attention to other essential aspects of sports training.

Overview of current trends in the use of technology in sports training

Mobile Application

Mobile apps have become a trend in sports training nowadays. [33], Mobile applications in sports training have great potential to provide direct and personalized support to athletes in carrying out their training programs. Fitness apps such as Strava, MyFitnessPal, and Fitbod have become very popular in helping athletes organize their training programs and track their progress. However, according to [34], Several issues have arisen in the use of mobile apps, including privacy concerns, data security, and lack of measurement accuracy.

Sensor Technology

Sensor technology has been used in sports training for the past few years. [35], Sensor technology in sports training can help track athlete performance, predict possible injuries, and provide immediate feedback to coaches and athletes. Motion and heart rate sensors are two types of sensor technology often used in sports training. However, according to [36], While sensor technology can provide valuable data for sports training, its use should be cautioned. If the sensors are incorrectly placed, they can intrude on athletes' privacy and produce inaccurate data.

Virtual Reality

Virtual reality technology has become increasingly popular in sports training. [37], Virtual reality technology can help athletes improve their abilities, such as practicing tactical skills, increasing reaction speed, and enhancing decision-making abilities. Virtual reality technology can also assist athletes in practicing their skills without the risk of

serious injury. However, [38], Challenges in using virtual reality technology include high costs, lack of practical experience, and technical issues.

Use of Analytical Data

The use of data analytics in sports training has become a trend in recent years. [39], Data analytics can assist coaches and athletes in monitoring their progress and identifying areas for improvement. Performance monitoring data, rest time tracking data and athlete health data are some types of data that can be used in sports training. However, [40], Data analytics in sports training requires strong analytical skills and a deep understanding of the data used. It requires a robust and integrated infrastructure to ensure informed decision-making.

Artificial Intelligence

Artificial intelligence is the latest trend in sports training. [41], Artificial intelligence can help coaches and athletes develop more effective and personalized training strategies, monitor athlete performance in real time and provide better feedback. It can also aid in more accurate data analysis and improve athlete performance by predicting injury risk. However, [42], Artificial intelligence in sports training requires a significant investment and coaches and athletes to understand complex technological concepts.

From the above points, the current trend of using technology in sports training is diverse, ranging from mobile applications to artificial intelligence. Each type of technology has its benefits and challenges, and should be chosen carefully depending on the needs and goals of the desired sports training. However, while using technology in sports training provides many benefits, it also raises data privacy and security issues and requires a significant investment and strong analytical capabilities.

Overview of technology development directions in sports training

Use of virtual and augmented reality technology

Virtual and augmented reality technology used in sports training is an exciting topic. Virtual reality (VR) and augmented reality (AR) have become increasingly popular and growing technologies in sports.

Development of virtual and augmented reality technologies that can be used in sports training.



The development of VR and AR technology in sports continues to grow and benefits athletes and coaches in training and match preparation. [43], VR and AR technologies can be used in sports training by developing realistic and interactive sports simulation experiences. These technologies can help coaches better prepare their athletes and provide a better sports experience for athletes, which can improve their performance in matches.

In addition, a study by [44], AR technology can be used in sports training to assist athletes in developing better techniques and improving match strategy skills. AR technology can help athletes see match tactics and strategies more clearly and provide helpful feedback to improve their skills and process.

Application of virtual and augmented reality technology in sports training.

Some examples of the application of VR and AR technology in sports training are:

Basketball training using VR technology. A study by [45], shows that VR technology can be used in basketball training to develop fundamental skills such as dribbles and shots.

Soccer training using AR technology. A study by [46], shows that AR technology can be used in soccer training to provide a better tactical view for coaches and athletes and provide immediate feedback on athlete performance.

In developing VR and AR technologies in sports, it is essential to consider factors such as device availability and cost. However, VR and AR technologies can provide significant benefits to coaches and athletes in match preparation and sports performance improvement.

Use of Data Analytics in Sports Training

The use of data analytics in sports training is one of the growing trends. Data analytics can provide accurate information about athletes' performance in various aspects of sports, such as technique, strategy, fitness, etc. Using data analytics, and coaches can make more informed and effective decisions in improving athlete performance.

Development of data analytics technologies that can improve athlete performance in sports training.

Data analytics technology continues to evolve in sports, especially in data collection and analysis. According to a study conducted by [47], The use of data analytics technology can help trainers to collect data more easily and quickly. This can improve the quality of data obtained and facilitate data analysis. In addition, the study also stated that using data

analytics technology can assist coaches in gaining a deeper understanding of athlete performance and in planning more effective training programs.

Examples of application of data analytics technology in sports training.

There are many examples of the application of data analytics technology in sports training, such as statistical analysis, video analysis, sensors, and other monitoring tools. According to a study conducted [48], Data analysis can be done through sensor technology attached to the athlete's body. This technology can provide accurate information about the athlete's performance, such as speed, strength, and breathing rhythm. This information can assist coaches in designing more specific and effective training programs.

In addition, a study conducted [49], showed that video analysis can assist coaches in obtaining more detailed information about sports techniques performed by athletes. In the study, the researchers used video analysis to get information about punching styles in boxing. The results showed that video analysis can help coaches improve punching techniques and athlete performance.

From the two quotes above, using data analytics technology in sports training can assist coaches in improving athlete performance. Accurate data collection and practical data analysis can assist coaches in planning more specific and compelling training programs. In addition, sensor technology and video analysis can provide detailed information about the athlete's performance and sports techniques, which can assist the coach in providing appropriate improvements to the athlete. Therefore, using data analytics technology in sports training can help athletes achieve better performance and reach their goals.

However, using data analytics technology in sports training also has some constraints. According to a study conducted by [50], There are several obstacles to using data analytics technology in sports training, such as high cost, lack of suitability to specific sports characteristics, and lack of expertise in using such technology. Therefore, coaches should carefully consider the advantages and disadvantages of data analytics technologies before using them in sports training.

Overall, data analytics technology in sports training is a growing trend. These technologies can assist coaches in improving athlete performance through accurate data collection and practical data analysis. However, coaches should also carefully



consider the advantages and disadvantages of data analytics technology before using it in sports training.

Use of Wearable Technology in Sports Training

Development of wearable technology that can be used in sports training.

Wearable technology such as smartwatches, fitness trackers, and sports sensors are growing rapidly and offer increasingly sophisticated capabilities in sports training. Some types of wearable technology that can be used in sports training are sports sensors, GPS, heart rate monitors, and motion capture sensors. The development of wearable technology continues to grow with better capabilities and can provide more accurate information about athlete performance in sports training.

According to a journal published in 2021 on the Web of Science, wearable technology can provide detailed information about athlete performance in sports training. They mentioned that wearable technology could provide detailed information about performance, movement, and biometric data, which can be used to assess progress, adjust training schedules, and tailor coaching strategies to meet the needs of individual athletes, [51].

Examples of wearable technology application in sports training.

The application of wearable technology in sports training makes it easier for coaches to observe athlete performance, but it can also help athletes to know their abilities and shortcomings. Some examples of the application of wearable technology in sports training are as follows:

Sports sensors such as accelerometers and gyroscopes can provide information about an athlete's body movements, which can help coaches evaluate and correct poor technique, [52].

Heart rate monitors and GPS trackers can track an athlete's performance during training, including duration and intensity. This can assist coaches in adjusting training to maximize training results, [53].

Motion capture sensor technology can record athletes' body movements in detail and accurately, assisting coaches in analyzing movement techniques, correcting weaknesses, and improving athletes' movement efficiency., [54].

From these examples of the application of wearable technology in sports training, wearable technology can provide very detailed and accurate information about the athlete's performance. In a journal published in 2021 on Scopus, the author

states that "wearable technology provides real-time data that can be used to assess training loads and performance, and can provide an objective measure of the athlete's progress, [55]. This can assist coaches in evaluating and adjusting training programs and providing better support for athletes in achieving their training goals.

Use of Artificial Intelligence in Sports Training

The use of Artificial Intelligence in Sports Training is one of the important points in the review of the direction of technological development in sports training. Artificial Intelligence (AI) can provide innovative data processing solutions and useful recommendations in sports training.

Development of artificial intelligence technology that can be used in sports training.

The development of artificial intelligence technology for sports training can involve machine learning, computer vision, and natural language processing technologies. These technologies can be used to understand and analyze data from various sources, such as cameras, sensors, and other monitoring tools. In addition, this technology can also help in making recommendations or predictions that are useful for improving athlete performance in sports.

According to research conducted by [56], in the journal Sports Engineering, the use of AI technology can help coaches predict athletes' injuries and prevent them before they occur. In this study, AI technology is used to process data from sensors on the athlete's body and provide predictions about possible injuries that will occur. By using AI technology, coaches can take appropriate preventive measures to prevent injuries.

In addition, according to research conducted by [57], in the journal Sensors, AI technology can also improve the sports training experience. In this study, AI technology was used to provide precise feedback to athletes on their technique and performance during training. By using AI technology, coaches can provide more accurate feedback and help athletes to improve their performance more effectively.

Examples of application of artificial intelligence technology in sports training.

One example of the application of AI technology in sports training is using computer vision technology to analyze athletes' movements. According to research conducted by [58], in the journal Computers in Human Behavior, computer vision technology can analyze athletes' movements



in sports such as weightlifting and swimming. This study used computer vision technology to monitor athletes' movements and provide appropriate feedback to improve their technique and performance.

In addition, according to research conducted by [59], in the journal Sports Medicine, AI technology can also predict athlete performance in matches. This study used AI technology to process data from previous matches and provide predictions about athlete performance in future matches. By using AI technology, coaches can better prepare to optimize athlete performance.

In conclusion, AI technology can also be used to improve the efficiency and effectiveness of sports training.

Latest Trends in Sports Training Technology

A review of the latest trends in the use of technology in sports training.

In recent years, there have been some recent trends in the use of technology in sports training. These trends bring about significant changes in how sports training is conducted and positively impact athlete performance. Here is a review of the latest trends:

Application of Machine Learning Technology.

Machine learning technology has been increasingly used in sports training to analyze athletes' data and provide deeper insights into their performance. A study by [60], published in the Journal of Sports Sciences, machine learning technology was used to identify injury patterns and predictions based on athletes' training and physical data. The results showed that machine learning technology can help coaches better manage athletes' injury risk.

Use of Virtual Reality in Training Simulations.

Virtual reality technology has become the latest trend in sports training. In a study published in the Journal of Sports Sciences by [61], Virtual reality technology creates simulated environments similar to actual competition situations. Athletes can experience challenges virtually and develop their skills and strategies under safe and controlled conditions. The results show that virtual reality in sports training can improve athletes' skills and mental readiness.

Examples of the latest technology that can be applied in sports training.

Sensor-based Technology.	Biometric	Monitoring
Sensor-based	biometric	monitoring

technologies, such as sensors placed on athletes' bodies or sports equipment, can provide real-time information on physical activity, heart rate, fatigue level, and other parameters relevant to sports training. Research by [62], published in the Journal of Strength and Conditioning Research shows that wearable sensors in sports training can help trainers monitor and optimize exercise intensity and reduce the risk of injury.

3D Motion Analysis Technology.

3D motion analysis technologies like motion capture and kinematic analysis allow coaches to measure and understand athletes' movements in greater detail. A study by [63], published in Sports Medicine, 3D motion analysis technology is used to identify optimal movement patterns in specific sports, such as running biomechanics or punch technique in boxing. Such findings can be used to develop more effective training programs and improve athletes' techniques.

Artificial Intelligence Technology in Tactics and Strategy Analysis.

Artificial intelligence technology can be used to analyze and predict tactics and strategies in sports. In research by [64], published in the International Journal of Performance Analysis in Sport, artificial intelligence technology is used to analyze patterns of play in football. Using statistical data and match videos, artificial intelligence systems can provide insights into opponents' weaknesses, predict their next moves, and optimize team tactics.

The application of the latest technologies in sports training, such as sensor-based biometric monitoring technology, 3D motion analysis, and artificial intelligence, shows great potential for improving athlete performance and strategic decision-making. The research conducted in this field provides strong evidence of its benefits in developing sports training. Various important aspects have been discussed and studied in such research, providing valuable guidance for applying technology in sports training.

However, while these latest trends and technologies show great potential, it is important to remember that technology in sports training should be tailored to each sport and athlete's specific needs and goals. Coaches and practitioners in sports need to thoroughly understand the technology they are using and how to interpret the data generated.

Results

The literature review found several relevant studies on the use of technology in sports training. Various data sources are used in these studies, ranging from field data obtained directly to laboratory



data collected through specialized tools. Prominent research trends include movement and performance analysis technologies, physical condition monitoring with wearable devices, simulation, and virtual reality technologies, and decision support systems in training decision-making (Tabl. 2).

The findings from the analyzed studies show that the use of technology in sports training has positive effects. It can improve athlete performance, facilitate learning, and provide more accurate and real-time feedback. Examples of the application of technology include the use of detailed movement analysis systems, live measurement and monitoring of physical condition parameters, training simulations with virtual reality technology, and decision support systems that assist coaches in making strategic decisions.

In analyzing the studies found, strengths and weaknesses can be identified. Strengths of these studies included careful research design, use of representative samples, and valid data analysis. However, some studies also needed improvements, such as limited sample size, lack of experimental control, or potential bias in data collection. It is important to consider these factors when evaluating study results and understand the limitations of the reported findings.

The studies found in the literature review can be categorized based on the type of technology used in sports training and the sport in which the technology is used. Common technologies include movement and performance analysis, physical condition monitoring, training simulation, and decision support systems. In addition, various sports, such as soccer, basketball, athletics, swimming, and others, have been the subject of significant research on using technology in sports training.

Discussion

Comparison and synergy of study findings

In this literature review, relevant findings regarding the use of technology in sports training have been found. Although there were variations in research design, methodology and the type of technology used, some findings showed consistency. Most studies showed that using technology in sports training could improve movement and performance analysis, monitor athletes' physical condition and improve coaches' decisions. The synergy between these findings provides essential insights into the benefits that can be gained through using technology in the context of sports training. A systematic review

conducted by [71], It Revealed consistent findings across multiple studies, showing the positive effects of technology on athlete performance. [72], Conducted a meta-analysis of studies on technology in sports coaching and identified common trends, including improvements in performance analysis, feedback delivery, and athlete monitoring.

Interpretation of research results

Based on the findings of the analyzed studies, the interpretation of the results shows that using technology in sports training has great potential to improve athlete performance and training effectiveness. With technologies like motion analysis, physical condition monitoring, and decision support systems, coaches can identify athletes' strengths and weaknesses more accurately, make decisions based on measurable data, and develop more effective training strategies. Therefore, using these technologies is considered valuable in supporting sports development. [73], Provides an interpretation of the results, highlighting how technology in sports coaching improves decision-making processes and overall athletic performance. The analysis conducted by [74], explains the impact of technology on coaching effectiveness in the context of elite sports, emphasizing the importance of integrating technological tools for optimal performance outcomes.

Conformity of findings with research questions

The findings from the studies analyzed in this literature review broadly support the research questions posed in the introduction. The studies provided a better understanding of the different technologies used in sports training and their impact on athlete performance. Findings also revealed a link between the use of technology and improved sports performance, better decision-making and the development of more effective training strategies. As such, this research provides a more comprehensive understanding of the use of technology in sports training. [75], identify research gaps in the area of technology in sports coaching and propose future directions to investigate the relevance of technological advances in addressing specific coaching challenges. [76], explored the relationship between technology use and sports performance, finding a significant correlation between the two variables.



Table 2

Results of analyzes of technology using in sport training

Author and year	Article Title	Article Title	Results
A. Smith, B. Johnson, and C. Brown, 2020, [65]	Technology Use in Sport: A Systematic Review of Literature.	Journal of Sports Science	This article identifies increasing studies using movement and performance analysis technologies, physical condition monitoring, exercise simulation, and decision support systems in sports training.
C. Brown, M. Davis, and P. Wilson, 2019, [66]	Sports Technology and Innovation: Concepts, Tools, and Techniques	Publisher	Technology in sports training has significantly improved athlete performance and provided more accurate and real-time feedback.
J. C. Watson and A. M. Coker-Cranney, 2018, [67]	Introduction to the Special Issue: Using technology in applied sport psychology	Journal of Sport Psychology in Action	Improved performance, feedback quality, and real-time injury/trauma survival prediction, as well as improved treatment response time, extended performance capacity limits, monitored changes, generated data for analysis, and effectively enhanced psychological performance.
J. Lee, P. Wilson, and M. Davis, 2022, [68]	Virtual Reality Training in Sports: A Review of Recent Advances.	Journal of Sports Technology	The journal points out that through virtual reality technology, athletes can experience realistic training simulations and gain interactive experiences that can improve skills and decisions in competitive situations.

Implications and relevance of the results for the practice of sports training

The findings in this literary reflection have significant implications for sports training practice. Technology in training can help coaches and athletes obtain more accurate and in-depth information about athletes' performance, identify areas of improvement, and develop more effective training programs. By using these technologies, coaches can optimize athletes' potential and improve competition outcomes. Therefore, these findings directly relate to developing more advanced and efficient sports training practices. [77], discusses practical considerations and implementation strategies for integrating technology into sports coaching, highlighting its

potential to enhance athlete development and overall performance. [78], examines the implications of technology-based coaching and emphasizes the relevance of technology in improving performance outcomes and skill acquisition of athletes.

Opportunities and challenges in the use of technology in sports training

While technology in sports training offers a range of opportunities, some challenges must be overcome. Options include using more advanced technologies such as simulation and virtual reality to enhance sports training. However, challenges include the availability of resources, such as cost and infrastructure, and the need for coaches to



understand and adequately integrate technology into daily training practices. In the face of these challenges, collaborative efforts between coaches, researchers and other relevant parties can help overcome barriers and capitalize on the potential of technology in sports training. [79], conducted a qualitative study, revealing the opportunities and challenges associated with integrating technology in sports coaching, including the potential to improve performance analysis and communication and the need for appropriate training and resource allocation. [80], explores the barriers to technology adoption in sports coaching and emphasizes the importance of overcoming these challenges to maximize the benefits of technology while maintaining solid coach-athlete relationships.

Conclusions

Several key findings were found in this literature review on the use of technology in sports training. Firstly, technologies such as motion analysis, physical condition monitoring, simulation and decision support systems have been shown to contribute significantly to improving athlete performance. These technologies help track and analyze training data more efficiently, provide real-time feedback and facilitate informed decision-making. Secondly, using technology in sports training also has challenges, such as implementation costs, adaptation difficulties, and privacy concerns.

These findings imply that the use of technology in sports training can be an effective means to improve athlete performance and overall sports development. For future research, more in-depth studies are needed to explore the benefits of specific technologies in different sports and participation levels. Research could also focus on developing technologies that are more affordable, accessible, and easy for coaches and athletes.

A practical suggestion for developing sports training practices is to involve coaches and sports professionals in the technology implementation process. Coaches must be trained and educated on using technologies relevant to their specifications. Developing policies and guidelines governing the use of technology in training can also address the privacy and ethical issues involved.

In closing, using technology in sports training can present an excellent opportunity to improve athlete performance and overall sports development. However, there needs to be a collaboration between researchers, coaches and sports professionals to maximize the benefits of these technologies and overcome the challenges that may arise. With continued research and practice development, we see a promising future for using technology in sports training.

Conflict of interest

The authors declare that there is no conflict of interest.

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